The documentation and process conversion measures necessary to comply with this revision shall be completed by 14 July 2000.

METRIC

MIL-PRF-19500/586E
14 April 2000
SUPERSEDING
MIL-PRF-19500/586D
10 July 1998

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, SCHOTTKY BARRIER, HERMETIC
TYPES 1N5819-1, 1N5819UR-1, 1N6650-1, 1N6650UR-1, 1N6761-1 AND 1N6761UR-1
JAN, JANTX, JANTXV, JANS, JANJ, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, Schottky barrier diodes. Five levels of product assurance are provided for each encapsulated device types as specified in MIL-PRF-19500, and two levels of product assurance for each unencapsulated device type die.

1.2 Physical dimensions. See figure 1 (DO-41), figure 2 (DO-213AB), figures 3 and 4 (JANC die) dimensions.

1.3 Maximum ratings.

Types	V_{RWM}	I_{O1} (3)	I_{FSM}	T_{STG}	T_J
	V (pk)	A dc	A (pk)	C°	C°
1N5819-1, 1N5819UR-1	45 (1)	1.0	25	-65°C to +150°C	-65°C to +125°C
1N6650-1, 1N6650UR-1	45(1)	1.0	25	-65°C to +150°C	-65°C to +150°C
1N6761-1, 1N6761UR-1	100 (2)	1.0	25	-65°C to +150°C	-65°C to +125°C

(1) Derate linearly at 1.8 V/°C above T_L or T_{EC} = +100°C for 1N5819-1, T_L or T_{EC} = +125°C for 1N6650-1 where T_L is at L = .375 inch.
(2) Derate linearly at 4.0 V/°C above T_L or T_{EC} = +100°C where T_L is at L = .375 inch.
(3) Derate linearly at 14 mA/°C above T_L or T_{EC} = +55°C for 1N5819-1 and 1N6761-1, T_L or T_{EC} = +80°C for 1N6650-1 where T_L is at L = .375 inch.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: (ex: Commander, Defense Supply Center, Columbus, ATTN: DSCC/VAC, Post Office Box 3990, Columbus, OH 43216-5000), by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Primary electrical characteristics. Unless otherwise specified, primary electrical characteristics at T_A = +25°C.

Types	Max V_{FM1} I_F = 0.1 A	V_{F2} I_F = 1.0 A	Max V_{FM3} I_F = 3.1 A	Max I_{RM} @ V_{RWM} pulsed method (see 4.5.1)		Max C_T V_R = 5 V dc	Max $R_{\theta JL}$ or $R_{\theta JEC}$ 3/8 lead length or end cap	Max $Z_{\theta JX}$
				T_J = +25°C I_{RM1}	T_J = +100°C I_{RM2}			
	V (pk)	V (pk)	V (pk)	mA	mA	pF	°C/W	°C/W
1N5819-1	0.34	0.49	0.80	0.05	5.0	70	70	12
1N5819UR-1	0.34	0.49	0.80	0.05	5.0	70	40	12
1N6650-1	0.39	0.56	0.80	0.025	1.0	70	70	12
1N6650UR-1	0.39	0.56	0.80	0.025	1.0	70	40	12
1N6761-1	0.38	0.69	NA	0.10	12.0	70	70	12
1N6761UR-1	0.38	0.69	NA	0.10	12.0	70	40	12

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

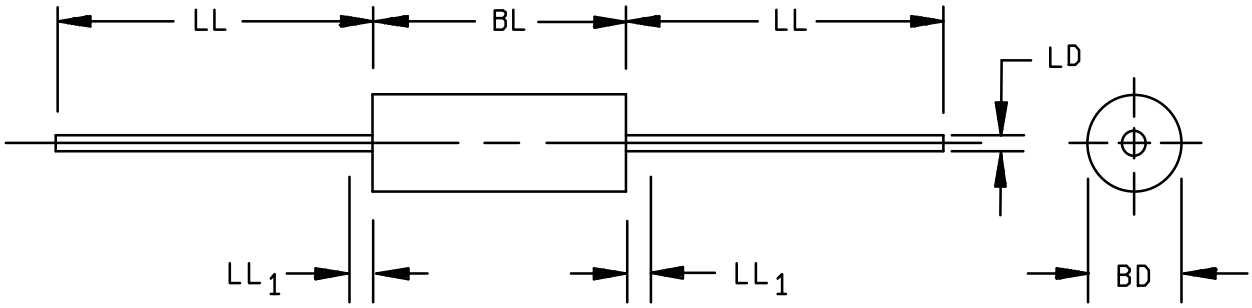
DEPARTMENT OF DEFENSE

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Services (DAPS), Building 4D (NPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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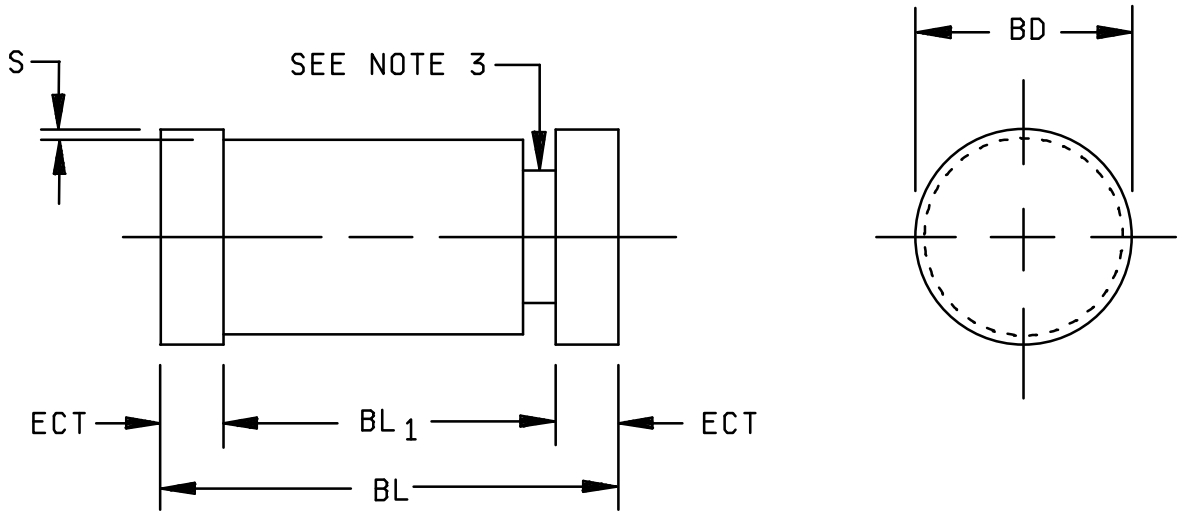


Symbol	Dimensions				Notes
	Millimeters		Inches		
	Min	Max	Min	Max	
LD	0.71	0.86	.028	.034	
BD	2.03	2.72	.080	.107	3
BL	4.06	5.21	.160	.205	3
LL	25.40		1.000		
LL_1		1.27		.050	4

NOTES:
1. Dimensions are in metric.
2. Inch pound equivalents are given for general information only.
3. Package contour optional within cylinder of diameter BD and length BL. Slugs if any shall not be included within this cylinder, but shall not be subject to the minimum limit of BD.
4. Lead diameter not controlled in this zone to allow for flash, lead finish build-up, and minor irregularities other than slugs.

FIGURE 1. Physical dimensions, 1N5819-1, 1N6650-1 and 1N6761-1 (DO-41).

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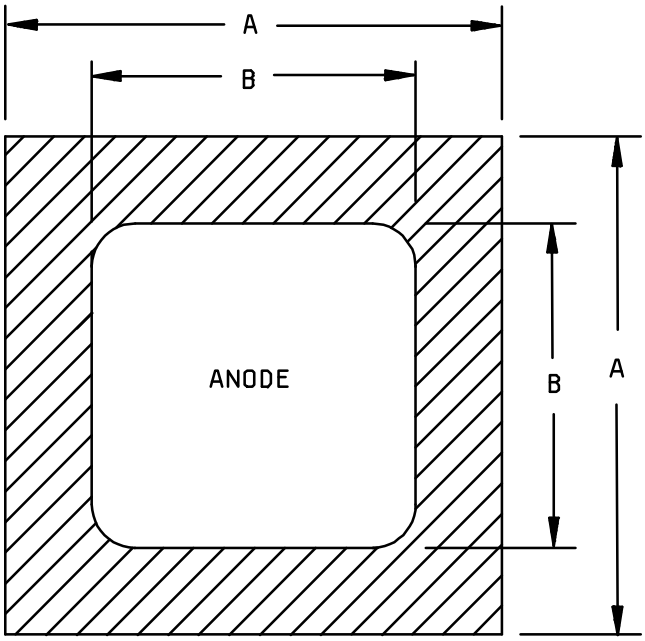


Symbol	Dimensions			
	Millimeters		Inches	
	Min	Max	Min	Max
BL	4.80	5.21	.189	.205
BL_1	4.04 REF		.159 REF	
BD	2.39	2.67	.094	.105
ECT	0.41	0.56	.016	.022
S	0.03		.001	

NOTES:
1. Dimensions are in metric.
2. Inch pounds equivalents are given for general information only.
3. Gap not controlled, shape of body and gap not controlled.

FIGURE 2. Physical dimensions of surface mount family, 1N5819UR-1, 1N6650UR-1 and 1N6761UR-1 (DO-213AB).

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BACKSIDE IS CATHODE

Symbol	Dimensions			
	Millimeters		Inches	
	Min	Max	Min	Max
A	0.89	0.99	.035	.039
B	0.79	0.84	.031	.033

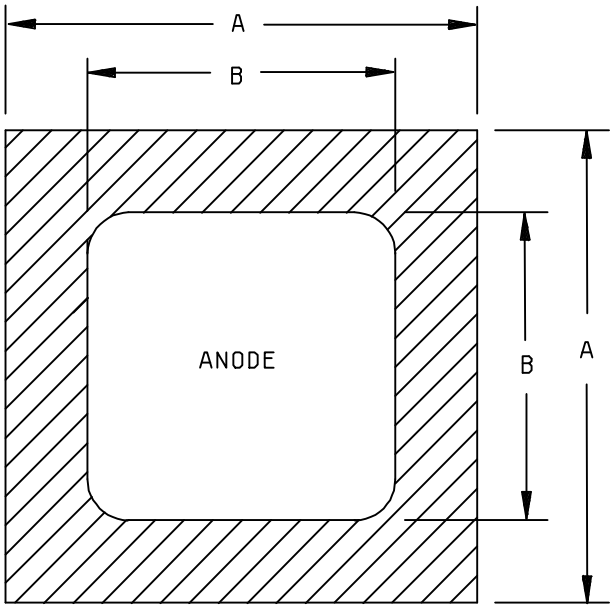
Design data

Metallization:
Top: (Anode) . . . . . AL
Back: (Cathode). . . . Au

AL thickness . . . . . 25000 Å min
Gold thickness . . . . . 4000 Å min
Chip thickness . . . . . 10 Mils ± 2 Mils

FIGURE 3. JANC (A-version) die dimensions.

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BACKSIDE IS CATHODE

Symbol	Dimensions			
	Millimeters		Inches	
	Min	Max	Min	Max
A	0.94	1.04	.037	.041
B	0.84	0.89	.033	.035

Design data

Metallization:
Top: (Anode) . . . . . AL
Back: (Cathode). . . . . Au

AL thickness . . . . . 20000 Å min
Gold thickness . . . . . 5000 Å min
Chip thickness . . . . . 12 Mils ± 2 Mils

FIGURE 4. JANC (B-version) die dimensions.

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3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-19500.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figures 1, 2, 3, and 4 herein.

3.4.1 Lead finish. Unless otherwise specified, lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Lead material. Lead material shall be copper clad steel with a minimum of 50% copper by weight.

3.5 Diode construction. These devices shall be constructed in a manner and using materials which enable the diodes to meet the applicable requirements of MIL-PRF-19500 and this document.

3.5.1 Surface mount. The U version shall be considered structurally identical to the non-U version except for lead attach.

3.6 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6.1 Marking for UR devices. UR-suffix devices only, all marking (except 3.6.2 below) may be omitted from the body, but shall be retained on the initial container.

3.6.2 Polarity. The polarity of all types shall be indicated with a contrasting color band to denote the cathode end. Alternatively, for UR suffix devices, a minimum of three contrasting color dots spaced around the periphery on the cathode end may be used.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3.

3.8 Electrical test requirements. The electrical test requirements shall be specified in group A, table I.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

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4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 JANHC and JANKC devices. Qualification for devices shall be in accordance with MIL-PRF-19500. This testing may be performed on a TO-5 package in lieu of the DO-41 axial leaded package.

4.2.2 JANJ devices. For JANJ level, 3.3.1 through 3.3.1.3 of MIL-PRF-19500 shall apply, except as modified herein. Supplier imposed requirements as well as alternate screens, procedures, and/or controls shall be documented in the QM plan and must be submitted to the Qualifying Activity for approval. When alternate screens, procedures, and/or controls are used, in lieu of the JANJ screens herein equivalency shall be proven and documented in the QM Plan. Radiation characterization may be submitted in the QM plan at the option of the manufacturer, however, paragraph 3.3.1.1 of MIL-PRF-19500 is not required. Die lot control and rework shall be in accordance with MIL-PRF-19500 paragraph 3.13 and D 3.13.2.1 for JANS level. Lot formation and conformance inspection requirements for JANJ shall be those used for JANTXV devices as a minimum.

4.2.3 JANJ Qualification. For JANJ qualification, 4.4.2.1 herein shall be performed as required by the Qualifying Activity.

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4.3 Screening (JAN, JANTX, JANTXV, JANJ, and JANS levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS Level	JANJ Level
1a 1b	Required Required	Not Required Required
2	Not Required	Not Required
3a 3b 3c	Required Not Applicable Required (see 4.5.4)	Required Not Applicable Required (see 4.5.4)
4, 5, 6 and 7a	Not Applicable	Not Applicable
7b	Required	Required
8	Required	Not Required
9	Required I_{R1} and V_{F2}	Required I_{R1} and V_{F2}
10 (2)	Required T_A = +100°C; V_{RWM} = 100 V(pk); 1N6761; T_A = +110°C; V_{RWM} = 45 V(pk); 1N5819, 1N6650, I_O = 0, half sine wave, f = 60 Hz	Required T_A = +100°C; V_{RWM} = 100 V(pk); 1N6761; T_A = +110°C; V_{RWM} = 45 V(pk); 1N5819, 1N6650, I_O = 0, half sine wave, f = 60 Hz,
11	Required $\Delta I_{R1} \leq$ 100 percent of initial reading or .02 mA whichever is greater. $\Delta V_{F2} \leq \pm$ 50 mV dc.	Required $\Delta I_{R1} \leq$ 100 percent of initial reading or .02 mA whichever is greater. $\Delta V_{F2} \leq \pm$ 50 mV dc.
12	Required See 4.3.2	Required See 4.3.2, t = 240 hours min.
13	Required Subgroup 2 of table I herein; $\Delta I_{R1} \leq$ 100 percent of initial reading or .02 mA whichever is greater; $V_{F2} \leq \pm$ 50 mV dc.	Required Subgroup 2 of table I herein; $\Delta I_{R1} \leq$ 100 percent of initial reading or .02 mA whichever is greater; $V_{F2} \leq \pm$ 50 mV dc.
14a 14b	Not Applicable Optional (3)	Not Applicable Optional (3)
15	Required	Required. Attributes Data only. Film or Non-Film Techniques may be utilized
16	Required	Required
17	Not applicable	Required Subgroup 2 of table I herein

See notes at end of table.

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4.3 Screening (JAN, JANTX, JANTXV, JANJ, and JANS levels only) Continued.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANTXV and JANTX Level	JAN level
1a 1b	Not Required Required (JANTXV only)	Not Required Not Required
2	Not Required	Not Required
3a 3b 3c (1)	Required Not Applicable Required (see 4.5.4)	Required in accordance with MIL-PRF-19500, JANTX level Not Applicable Required (see 4.5.4)
4, 5, 6 and 7a	Not Applicable	Not Applicable
7b	Required	Not Required
8	Not Required	Not Required
9	Not Applicable	Not Applicable
10 (2)	Required T_A = +100°C; V_{RWM} = 100 V(pk); 1N6761; T_A = +110°C; V_{RWM} = 45 V(pk); 1N5819, 1N6650, I_O = 0, half sine wave, f = 60 Hz	Not Applicable
11	Required I_{R1} and V_{FM2}	Not Applicable
12	Required See 4.3.2, t = 48 hours	Not Applicable
13	Required Subgroup 2 of table I herein; $\Delta I_{R1} \leq$ 100 percent of initial reading or .02 mA whichever is greater; $V_{F2} \leq \pm$ 50 mV dc.	Not Applicable
14a 14b	Not Applicable Optional (3)	Not Applicable Not Required
15 and 16	Not Required	Not Required
17	Not Applicable	Not Applicable

(1) Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with table IV of MIL-PRF-19500, screen 3 prior to this thermal test.
(2) Junction temperature (T_J) is not to exceed 115°C at V_{RWM}. T_J is affected by the device mounting thermal resistance when parasitic power is generated by the temperature dependent leakage current. Until this leakage becomes significant near thermal runaway, T_J remains approximately equal to T_A or T_J for I_O = 0.
(3) Per MIL-PRF-19500

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4.3.1 Screening (JANHC or JANKC). Screening of die shall be in accordance with Discrete Semiconductor Die/Chip Lot Acceptance appendix of MIL-PRF-19500 and die shall be 100 percent probed in accordance with group A, subgroup 2.

4.3.2 Burn-in conditions. Burn-in conditions are as follows: I_F = 1.0 A dc (min), T_A = room ambient as defined in the general requirements of MIL-STD-750. Mounting and test conditions in accordance with MIL-STD-750, method 1038, test condition B.

4.3.2.1 Mounting. Devices may be mounted using any convenient method including the temporary attachment of leads on UR suffix devices, provided that the parts are burned-in at an average lot T_J = 110 ±15°C.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table V of MIL-PRF-19500, and table I herein. The following test conditions shall be used for $Z_{\theta JX}$, group A inspection:

a.	I_M measurement current	1 mA to 10 mA.
b.	I_H forward heating current	2.0 A.
c.	t_H heating time	10 ms.
d.	t_{MD} measurement delay time	70 µs maximum.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	4066	I_{FSM} = 25 A(pk), condition A 2, I_O = 1.0 A; T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); 5 surges of 8.3 ms each at 1 minute intervals.
B4	1036	I_F = 1.0 A; T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles.
B5	1038	T_A = +100°C minimum; V_R = V_{RWM}, I_O = 0, half sine wave, f = 60 Hz. T_A adjusted as required to achieve a T_J = +110°C minimum.
B6	3101	$R_{\theta JL}$ = 70°C/W; L = .375 inch (9.52 mm) lead length (non-surface mount). $R_{\theta JEC}$ = 40°C/W (surface mount).

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4.4.2.2 Group B inspection, table VIb (JAN, JANTX, JANTXV, AND JANJ of MIL-PRF-19500).

Subgroup	Method	Condition
B2	4066	I_{FSM} = 25 A(pk), condition A 2, I_O = 1.0 A; T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); 5 surges of 8.3 ms each at 1 minute intervals.
B3	1038	T_A = +100°C minimum; V_R = V_{RWM}, I_O = 0, half sine wave, f = 60 Hz. T_A adjusted to achieve T_J =+110°C minimum.
B4	2075	In accordance with 4.5.2.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.

4.4.3.1 Group C inspection, table VII of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Tension: Test condition A; weight = 10 pounds; t = 15 s. Lead fatigue: Test condition E; weight 1 pounds. NOTE: Both tension and lead fatigue are not applicable for UR devices.
C6	1038	T_A = +100°C minimum; V_R = V_{RWM}, I_O = 0, half sine wave, f = 60 Hz. T_A adjusted to achieve T_J =+110°C minimum.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions for subgroup testing in table IX of MIL-PRF-19500 and table II herein. Electrical measurements (end- points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in 4.3.2.1 of MIL-STD-750.

4.5.2 Decap internal visual scribe and break (DPA). Scratch glass at cavity area with diamond scribe. Carefully snap open. Using 30X magnification, examine the area where die or bond material were in contact with the plug and verify metallurgical bonding area.

4.5.3 Steady-state operation life. This test shall be conducted with a half-sine wave of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall not be greater than 180° nor less than 150°.

4.5.4 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum screen limit shall be developed by the supplier using statistical methods and it shall not to exceed the Group A, Subgroup 2 herein.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance	3101	See 4.5.4	$Z_{\theta JX}$		12	°C/W
Forward voltage	4011	I_F = .1 A (pk) pulsed (see 4.5.1)	V_{FM1}			V
1N5819-1, 1N5819UR-1					0.34	
1N6650-1, 1N6650UR-1					0.39	
1N6761-1, 1N6761UR-1					0.38	
Forward voltage	4011	I_F = 1.0 A (pk) pulsed (see 4.5.1)	V_{F2}			V
1N5819-1, 1N5819UR-1					0.49	
1N6650-1, 1N6650UR-1					0.56	
1N6761-1, 1N6761UR-1					0.69	
Forward voltage 1N5819-1, 1N5819UR-1 1N6650-1, 1N6650UR-1	4011	I_{FM} = 3.1 A (pk) pulsed (see 4.5.1)	V_{FM3}		0.80	V
Reverse current leakage	4016	Pulse method	I_{RM1}			mA
1N5819-1, 1N5819UR-1		V_{RM} = 45 V (pk)			0.05	
1N6650-1, 1N6650UR-1		V_{RM} = 45 V (pk)			0.025	
1N6761-1, 1N6761UR-1		V_{RM} = 100 V (pk)			0.10	
Subgroup 3						
High temperature operation:		T_A = +100°C				
Reverse current leakage	4016	Pulse method (see 4.5.1)	I_{RM2}			
1N5819-1, 1N5819UR-1		V_{RM} = 45 V (pk)			5.0	mA
1N6650-1, 1N6650UR-1		V_{RM} = 45 V (pk)			1.0	mA
1N6761-1, 1N6761UR-1		V_{RM} = 100 V (pk)			12.0	mA
1N5819-1, 1N5819UR-1		V_R = 40 V (pk)	I_{RM3}		4.5	mA
1N6650-1, 1N6650UR-1		V_R = 40 V (pk)			0.7	mA
1N5819-1, 1N5819UR-1		V_R = 35 V (pk)	I_{RM4}		3.0	mA
1N6650-1, 1N6650UR-1		V_R = 35 V (pk)			0.5	mA

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						
Forward voltage	4011	Pulse method (see 4.5.1) I_F = 1.0 A (pk)	V_{F4}			V
1N5819-1, 1N5819UR-1					0.45	
1N6650-1, 1N6650UR-1					0.51	
1N6761-1, 1N6761UR-1					0.64	
Low temperature operation:		T_A = -55°C				
Reverse current leakage	4016	Pulse method (see 4.5.1)	I_{R5}			mA
1N5819-1, 1N5819UR-1		V_{RM} = 45 V (pk)			0.2	
1N6650-1, 1N6650UR-1		V_{RM} = 45 V (pk)			0.1	
1N6761-1, 1N6761UR-1		V_R = 90 V (pk)			0.4	
Forward voltage	4011	Pulse method (see 4.5.1) I_F = 1.0 A (pk)	V_{F5}			V
1N5819-1, 1N5819UR-1					0.65	
1N6650-1, 1N6650UR-1					0.65	
1N6761-1, 1N6761UR-1					0.80	
Subgroup 4						
Capacitance	4001	V_R = 5 V dc, .01 ≤ f ≤ 1 MHz, V_{SIG} = 15 mV p-p	C_T		70	pF

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Group E inspection (all quality levels) for qualification only. 1/

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			22 devices c = 0
Thermal shock (temperature cycling)	1051	500 cycles	
Hermetic seal	1071	Test condition E	
Electrical measurement		See table I, group A, subgroup 2	
Subgroup 2			22 devices c = 0
Steady-state reverse bias	1038	Test condition A; 1,000 hours, See C6 herein.	
Electrical measurement		See table I, group A, subgroup 2	
Subgroup 3			
Not applicable			
Subgroup 4			22 devices c = 0
Thermal resistance, (forward voltage drop diode method)	4081	$R_{\theta JL}$ = 70°C/W maximum at .375" lead length; $R_{\theta JEC}$ = 40°C/W max.; method 3101 per 4.5.4 except I_H = 1.0 A and T_H = 25 sec (min)	
Subgroup 5			
Not applicable			
Subgroup 6			
Not applicable			

1/ For initial design and process change verification only (one time testing).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML 19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from : Defense Supply Center, Columbus, ATTN: DSCC-VQE, P.O. Box 3990, Columbus, Ohio, 43216-5000.

6.4 Suppliers of die. The qualified die suppliers with the applicable letter version (example JANHCA1N5819) will be identified on the QML.

JANC ordering information		
	Manufacturer	
PIN	55801	59377
1N5819	JANHCA1N5819 JANKCA1N5819	JANHCB1N5819 JANKCB1N5819
1N6650	JANHCA1N6650 JANKCA1N6650	JANHCB1N6650 JANKCB1N6650
1N6761	JANHCA1N6761 JANKCA1N6761	JANHCB1N6761 JANKCB1N6761

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5961-2024)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/586E	2. DOCUMENT DATE 14 April 2000

3. **DOCUMENT TITLE** SEMICONDUCTOR DEVICE, DIODE, SILICON, SCHOTTKY BARRIER, HERMETIC TYPES 1N5819-1, 1N5819UR-1, 1N6650-1, 1N6650UR-1, 1N6761-1 AND 1N6761UR-1 JAN, JANTX, JANTXV, JANS, JANJ, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial 614-692-0510	DSN 850-0510	FAX 614-692-6939	EMAIL alan_barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad Street, Columbus, OH 43213-1199	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533, Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888			

DD Form 1426, Feb 1999 (EG) Previous editions are obsolete WHS/DIOR, Feb 99

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